AVOID THE TURBULENCE!

The definitive guide to top landing on the Long Mynd

For paragliders, hang gliders and rigid wings

Adapted from a 2021 Skywings article by club member Steve Young

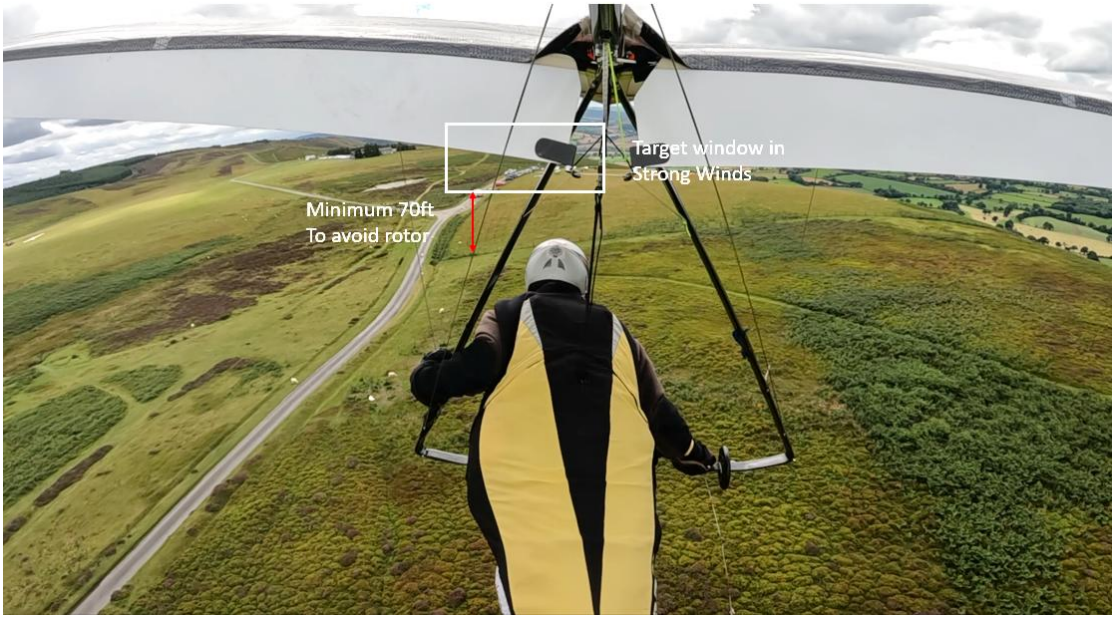
When planning a top landing, many considerations flash through a pilot's mind within just a few seconds — and some of them are far from obvious to the less experienced flyer. Three factors matter above all others:

1. Wind direction. Is the wind slightly off to one side or the other? Choosing a direction of approach that minimises the extent of the final turn into wind is important, as is a careful scan of the field for obstacles — overflying other gliders, spectators or parked cars should always be avoided.

2. The shape of the top of the hill. Understanding the terrain is essential. Any downslope component to a top landing makes it significantly more demanding, and on a hill like the Long Mynd the curved summit profile adds further complications, as we will see.

3. Wind speed. Wind speed governs the angle of your final glide and, combined with airspeed, dictates where you will touch down. Getting the height of your approach right is everything. This is the factor that catches pilots out most often.

Paragliders, hang gliders and rigid wings all share the same fundamental challenge: judging the right height at which to begin the approach. Come in too high and you may find yourself spending too long in the zone of accelerated airflow caused by the venturi effect over the hill — at risk of being blown back or of overshooting the field entirely. Come in too low and the approach will involve a high-speed, possibly high-banked turn close to the ground, where options may be dangerously limited.



View from a rigid wing glider showing how small the target landing window can be for faster wings.

The shape of the hill

At the Long Mynd, all of this is further complicated by the shape of the hill itself. The summit ridge is curved, almost like an aerofoil — and that profile has a profound influence on the local airflow. In light winds there is no great issue: the air simply follows the contours of the hill and turbulence is minimal. But as wind speed approaches 15 mph and above, the character of the airflow over the highest part of the ridge starts to change in ways that can catch out even the most experienced pilots.

At the Mynd, the change is easy to detect at 'Upright Corner'. Standing at the north-east corner of the top landing field on a strong wind day, two distinct effects make themselves felt: the wind speed seems greatly reduced, and what remains is gusty and variable — with sudden, unpredictable changes in both speed and direction.

View of the hill showing the position of the turbulent zone behind the crest to the West

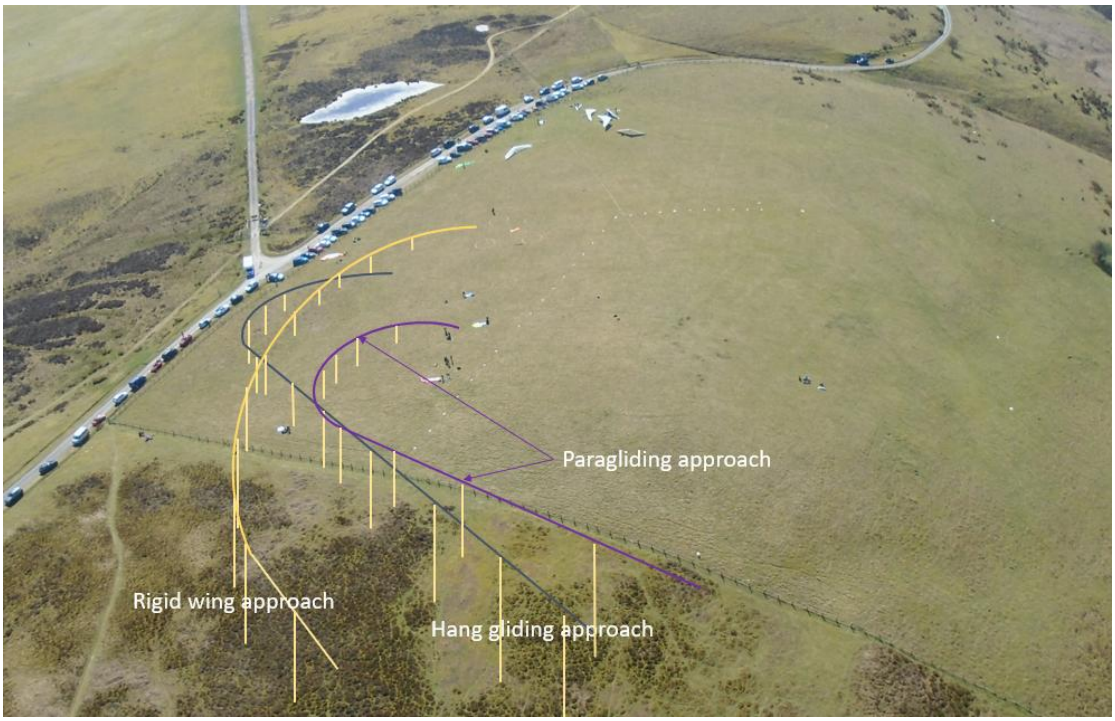
This behaviour is caused by the airflow separating from the surface of the hill as it crosses the crest. The mass of air has been rising up the front face at roughly 40 degrees to the horizontal, but as it passes over the summit it detaches from the ground and spills into the lee, generating a rotor and turbulence zone that can extend at least 30 feet above the rear corner of the field. Moving forward towards the crest reduces the effect considerably — which is why the area just behind the ridge line, east of the hang glider rigging area, is the ideal spot to aim for.

All three aircraft types can be caught out by this turbulence. Paragliders will often choose not to fly once wind speeds reach 15 mph, but conditions can deteriorate quickly and without warning. A paraglider does have the advantage that the wing sits further from the ground — and from the worst of the rotor — but an approach that runs too far back into the corner of the field can still result in a partial collapse at exactly the wrong moment.

Hang gliders and rigid wings must come in fast in windier conditions to cut through the wind gradient, which makes precise positioning absolutely critical. Too high and there is a risk of overshooting — not necessarily a problem if the overshoot area is clear. Too low and the wing enters the rotor just as the pilot commits to the final turn into wind. There is an additional and easily underestimated hazard here: on a lower, later turn, the downwind wing is passing through slower-moving air due to the wind gradient. This robs it of lift, making it genuinely difficult to bring the glider around into wind — not a situation anyone wants to be in at low altitude.

Choosing the approach

Given everything described above, the key principle is straightforward: in stronger wind conditions, a longer approach is almost always better. By setting up earlier and at greater height — closer to the technique that rigid wing pilots typically use — a pilot can position the aircraft well above and ahead of the turbulence at the corner of the field. From there, the turn into wind is gentler, the required bank angle is manageable, and the wing can be kept close to horizontal while aiming for the ground just behind the crest — where the air is smoothest and the reward is greatest.



The three differing landing approach paths for rigid wings, paragliders and hang gliders.

The southern approach to the Long Mynd top landing field — used when the wind has a northerly component — deserves special mention. It requires flying along the line of the road and directly over parked cars before committing to a tight turn into wind. This leaves extremely little margin if anything goes wrong or the aircraft encounters unexpected sink at a critical moment; it is an approach to be treated with great respect.

Before you fly

The principles described here apply well beyond the Long Mynd. Many top landing sites across the UK and abroad share the same aerofoil-like ridge profile, and with it the same potential for rotor turbulence in the lee of the crest. If you are planning to top land somewhere on a stronger wind day, take a few minutes before launching to walk around the field holding a streamer, paying close attention to wind speed and direction at each corner. What you find may genuinely surprise you. That small investment of time costs nothing — and may save a great deal more.